

AI4PD: A Multi-Agent Clinical Web Portal for Querying the Unseen States of Patients with Parkinson’s Disease

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Abstract. Interpreting Parkinson’s disease means relating motor examinations, brain imaging, and between-visit measurements. Parkinson’s Progression Markers Initiative analyses estimate motor states and pathway integrity, resolve the more affected side, and test longitudinal claims. Their results appear in separate tables, so a clinician must recover when each result applies and what evidence supports it. No clinician-facing Parkinson’s tool we reviewed combines population analyses, motion visualization, observation timing, source-record provenance, and study-level validity findings for one patient. We present AI4PD, a multi-agent web portal in which each analysis is an agent, a software module behind one shared interface. Each agent returns participant-specific results from a model fitted to the cohort. Each query’s evidence anchor, a participant and a month, limits displayed observations to that month or earlier. Results carry dates, source records, available uncertainty, and source-study validation findings. A deployed portal shows schematic and wearable-derived motion on a reference body. A locally tested evidence service replays frozen laterality and motor-change outputs; motor-state and pathway modules await verified model files. Of 86 portal participants, 30 have results from both implemented analyses, and only three have both for the same analysis month. The gap between 30 and three shows why integration must preserve analysis dates alongside participant identity. A proposed extension is a multi-scale digital twin with motion and neural-circuit modules for stimulation what-if queries [5]. AI4PD is a research prototype for clinician-in-the-loop interpretation. It does not diagnose or prescribe; its clinical usefulness remains unevaluated.

Keywords: Parkinson’s disease · Clinical decision support · Multi-agent system · Time-anchored evidence · Patient-specific visualization · Evidence provenance

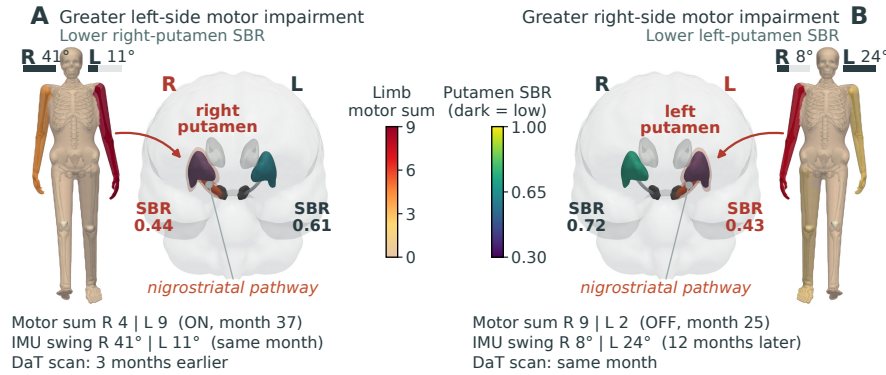


Fig. 1. Two PPMI participants show opposite patterns of asymmetry: greater motor impairment on one body side accompanies lower dopamine transporter (DaT) binding in the opposite putamen (values and acquisition gaps on the figure; months count from each participant’s first recorded examination). Bars are measured wearable (IMU) arm-swing amplitudes, limb tint is the paired MDS-UPDRS III item sum divided by 9, and SBR (specific binding ratio) is from the laterality study’s PPMI table. Anterior views face the participant (R on the viewer’s left). Tinted skin marks the worse limbs, the arrow leads to the contralateral putamen colored by measured SBR (darker is lower), and the lit tube is the nigrostriatal pathway, drawn schematically. Participant measurements are overlaid on reference anatomy (see credits) under a computed skin envelope. The pose is one frame of a separate WearGait-PD recording, mirrored for A. Per-person agreement under the sign convention of Sect. 3.4.

1 Introduction

Interpreting Parkinson’s disease (PD) means joining a motor examination, brain imaging, and between-visit measurements, each at its own interval. PD pathology is asymmetric: dopaminergic loss on one brain side drives signs on the opposite body side. PD is also heterogeneous: two patients with the same motor total can be in different motor states. The Parkinson’s Progression Markers Initiative (PPMI) [14] supplies participant records. BioFIND [12] and the Parkinson’s Disease Biomarkers Program (PDBP) are external evaluation cohorts for the source analyses, and the wearable WearGait-PD study [2] adds motion recordings. A growing number of such databases with population histories enables patient-specific, data-driven analysis. The analyses reviewed below recover these patterns and leave them in tables. A clinician will not read those tables during a visit. Reading them would mean matching participants, reconciling observation dates, and recovering each analysis’s validation limits. AI4PD brings participant-specific outputs into one workspace and preserves when each observation was taken, where each result came from, and the limits of its evidence.

Each piece of such a system already exists (Sect. 2.3), and no clinician-facing Parkinson’s tool we reviewed joins them for one patient at one point in time. A scan and an examination months apart are not one state. The more affected body

side suggested by imaging asymmetry can differ from the side identified by the motor examination. The workspace preserves this disagreement for inspection. A prediction validated in one cohort or enrollment era can fail in another.

AI4PD is our multi-agent web portal for that joining.⁴ It connects observed measurements with estimates of laterality, motor state, pathway integrity, and future motor change. Those estimates are the unseen states of the title, a latent space behind the records, and querying the unseen means retrieving one patient’s estimates with dates and evidence. Today the workspace retrieves laterality and motor-change results (Sect. 3.4). Each population analysis becomes one agent, and a portal shows its outputs for one patient. Every query is keyed to an *evidence anchor*, a participant and a historical month, and by default returns only observations dated on or before that month (Sect. 3). Each analysis uses patterns learned from cohort data to produce estimates for individual participants, a collaborative filtering against the cohort, not an average-cohort result. We make three contributions. (i) A shared interface tying each participant-specific result to its time, source records, uncertainty, and its source study’s validation findings (Sect. 3). (ii) A locally tested evidence workspace implementing that interface for laterality and motor-change outputs, alongside deployed motion-visualization clients (Sect. 4). (iii) An integration audit separating participant overlap from temporal alignment, with verification of retrieval fidelity and historical filtering (Sect. 5). Current users are research teams; no clinician study has been conducted.

2 Background and Prior Work

2.1 Source Analyses

Generic population models are fitted offline and produce participant-linked outputs, indexed by visit or forecast origin where applicable. The participant identifier links them, and each analysis becomes one back-end agent under the contract of Sect. 3.

Motor-state estimation. Tirhekar et al. ask which motor pattern describes a patient’s examination. The portal would show the likely state, and the uncertainty between states, that a total score hides. A Bayesian mixture model over five domains, selected from 2,912 configurations, groups 29,366 assessments from 1,847 PPMI participants into five states. Dopamine transporter (DaT) binding and FreeSurfer subcortical volumes differ across states. The fixed model assigns 99.7% of external BioFIND visits ($n = 310$) to a state with high model confidence, a measure of confidence rather than of accuracy against a clinical label [23].

Pathway-integrity profiling. Vinod et al. ask which brain circuits show reduced imaging integrity. The portal would show six integrity scores, one per circuit, on reference anatomy (Fig. 2). T1-weighted MRI volumes, free-water-corrected

⁴ Project page: <https://cvc-lab.github.io/projects/ai4pd/>

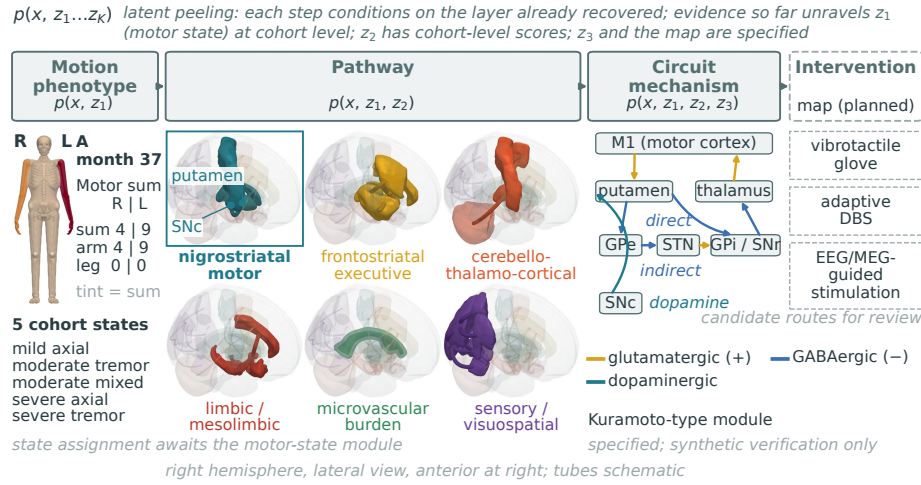


Fig. 2. Proposed progression from movement observations to pathway profiles, circuit models, and candidate interventions for one person. The top strip, latent peeling of $p(x, z_1 \dots z_K)$, is the design framework behind separately fitted modules: it would trace a patient back from observed measurements to progressively finer latent properties, each layer informed by the estimates of the layer before. Evidence so far unravels z_1 (motor state) at cohort level, z_2 has cohort-level scores, and z_3 and the map are specified. Motion phenotype: A’s four-limb burden on the Fig. 1 reference body (neutral pose) and the cohort’s five motor states [23]. Pathway: six circuits scored from DaT-SPECT, free-water diffusion and T1 MRI, one reference brain each (right hemisphere, A’s implicated side, lateral view, anterior at right; cerebellum and periventricular band are approximate atlas masks, and tubes are schematic curves between region centroids). Cohort-level nigrostriatal Multimodal Pathway Integrity Score correlation against the motor examination: $\rho = -0.20$ $[-0.30, -0.09]$, $n = 283$ [27]. Circuit mechanism: the basal-ganglia motor loop whose Kuramoto-type module is specified and verified on synthetic data only [5]. Intervention map (Sect. 6): future work listing candidate routes for clinician review.

diffusion measures, and DaT binding from 294 participants feed the six scores. Co-clustering of patients and features [26] is stable across random seeds and bootstrap resampling [27].

Striatal laterality. Tirhekar et al. ask whether imaging asymmetry matches the patient’s more affected side. The evidence workspace shows signed left–right putamen asymmetry beside each body side’s motor sum (Fig. 1). The sign keeps the direction that mean binding loses, and the pairing gives the scores anatomical meaning. The study recovers the contralateral relation of dopaminergic loss to motor burden in PPMI ($n = 1,367$) and an independent cohort ($n = 56$) [24].

Longitudinal prediction validity. Tirhekar et al. ask whether multimodal evidence supports a patient’s motor-change forecast. Seven validity gates, each with its

own verdict, test it, and the evidence workspace shows the held-out prediction with those verdicts. Across 5,404 cut-offs (forecast origins) from 1,058 participants, dated imaging and cognitive history add a small internal gain in predicting later motor change, and that gain fails the era and transfer gates [25].

2.2 The Gait Twin

The motion subsystem has two components: a view of PPMI gait metrics and WearGait-PD recordings [2] on an OpenSim musculoskeletal body model [19,20], and a separate mechanics engine that estimates joint moments from wearable data (inverse dynamics). Joint moments are the turning effects of forces at the joints, and the engine’s recoverability map states which of them each worn-sensor set (a sensor tier) can support. A reference-data consistency check, feeding the engine sensor-equivalent inputs derived from reference motion and center of pressure, reached a mean window-level ankle-moment correlation of $r = 0.995$ against a processed inverse-dynamics reference on AddBiomechanics walking subjects [28,8].

2.3 Related Systems

PD-Insighter combines patient-motion replay with a dashboard of movement measures and action labels [11], and Xu et al. link subject-level classifier predictions with anatomical fiber views [29]. Commercial DaT quantification reports side-resolved binding and a left-to-right ratio against a normal database, without the clinical motor side [15]. Digital twins for Parkinson’s score a new patient against a fixed population model [3,10], and SYMPHONY-DT gates each forecast on confidence [10]. None combines participant-specific outputs from several population analyses with motion visualization, observation timing, source-record traceability, and each analysis’s validation findings.

3 System Architecture

3.1 Overview

AI4PD has three tiers (Fig. 3). The *server* holds the cohort tables (the data spine), a versioned catalog built from them, and the evidence service. *Clients* are a deployed motion portal, a deployed gait twin explorer, and a locally tested evidence workspace, each opened for one patient. Between them sit per-analysis modules, called agents, linking the workspace to stored results. Each accepts a participant, a historical month (the evidence anchor), and a population-model version, then returns a finding, its source records, a claim status, and any available uncertainty estimate; a missing estimate is marked explicitly. One interface preserves how each source analysis defines and reports uncertainty. The implemented agents are deterministic adapters applying fixed rules to frozen, versioned outputs. None acts on its own or retrains online. The longer-term design

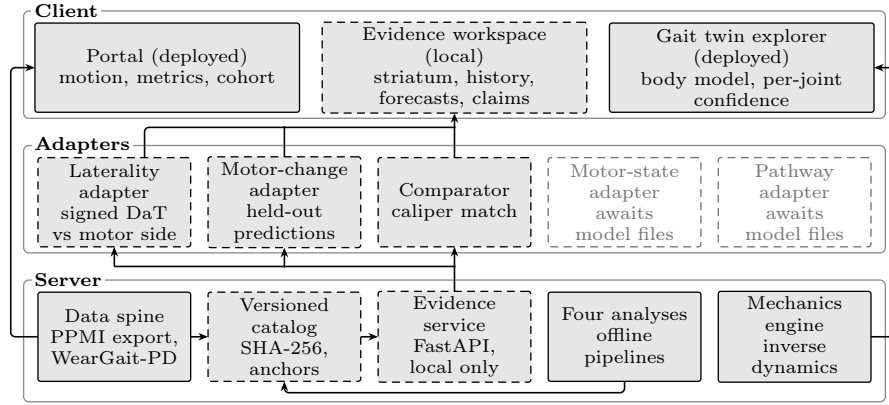


Fig. 3. AI4PD as built. Solid shaded boxes are deployed behind access control (portal, twin explorer) or run offline (data spine, four analyses, engine). Dashed shaded boxes are built and tested on a local host. Dashed gray boxes await verified model files.

links three scales per participant (Fig. 2, Sect. 6): a motion twin (macro), the neural-circuit twin (meso) of the neurophysiome deep brain stimulation (DBS) manuscript in preparation [5], and molecular context (micro). Each is a generative model fitted once and evaluated at query time. Linking records by participant and anchor is built; learning relationships across scales (multi-scale patient phenotyping) is the modeling goal.

3.2 Versioned Catalog

Each displayed result is traceable to a version of its source data and analysis. The data spine holds the WearGait-PD recordings, the frozen output tables of two analyses, and the PPMI export (examinations, imaging measures, demographics, dated assays). An offline builder reads these sources without modifying them and writes a catalog. SHA-256 fingerprints identify each source and analysis version (catalog `4ba1efb1`). Every value keeps its source file and row. A private map, never served, replaces PPMI identifiers with opaque ones.

3.3 The Evidence Anchor

The evidence anchor is a participant and a month counted from the participant’s earliest clinical-history, gait, or analysis month. Fig. 1 counts from the first examination; the two origins differ by any gap between first observation and first examination. By default the service returns only observations dated on or before that month, so a query at month 24 returns observations through month 24. A forecast available then may describe a later outcome. Each forecast has a forecast origin, the month from which its source study made it (the laterality study’s landmark, the motor-change study’s cut-off). The laterality adapter keeps its

source study’s rule: scan and examination within three months of each other and before the forecast origin. The motor-change adapter requires a participant and forecast-origin match. Observed follow-up appears only on request.

3.4 Analytical Modules

The evidence service implements adapters for striatal laterality and motor-change prediction. Motor-state and pathway-integrity adapters await verified model files. Imaging and examination measures share one sign convention: matching signs mean lower binding on one brain side and greater motor impairment on the opposite body side. For example, lower left-putamen binding and greater right-body impairment both give positive values. Let S be the putamen specific binding ratio (SBR) and U the sum of the paired motor-examination items for one body side, larger for greater impairment. Then $a_{\text{DaT}} = (S_R - S_L)/(S_R + S_L)$ and $a_{\text{motor}} = (U_R - U_L)/(U_R + U_L)$. Agreeing signs support a contralateral pattern without identifying a cause. The adapter returns the source study’s four ridge predictions of a_{motor} at 12 months, plus a separately fitted conformal predictor’s median and 90% interval, which belong to that predictor alone. The motor-change adapter returns four predictions of motor-score change in points per year, each made with that participant excluded from model fitting, with no individual uncertainty intervals. Their differences show only variation among models. A comparator returns up to three same-cohort participants within fixed tolerances (calipers) of motor total, mean putamen binding, and asymmetry, never using outcomes or clinical side. The matches are descriptive comparisons.

Cloudflare Workers host both deployed clients behind an access-controlled address without the raw wearable recordings; the FastAPI evidence service runs on a loopback address only with no route to raw sources or the identifier map.

4 Front-End and Back-End Integration

In the local evidence workspace a user selects a participant and a historical month, sees the examinations up to that month with the age of each scan and gait record, and traces every value to its source.

4.1 From Tables to Anchored Records

The portal export holds one summary record for each of 86 PPMI gait-substudy participants (49 PD, 37 prodromal). For the 78 with a motor examination it uses the latest one and selects the off-medication assessment when both medication states exist. Gait measures are linked within six months of that examination. Missing severity is left unknown. Dates and identifying fields are removed. The evidence workspace queries a separate catalog built from the same dated records, which keeps the observation dates and supports 997 participant-month queries for those 86 people. The 42 WearGait-PD recordings (PD and controls, unlinked to PPMI) become pose-per-frame clips.

4.2 Motion Visualization and Gait Twin

For PPMI participants, the portal generates a schematic animation from recorded gait metrics and examination scores on reference anatomy. Gait speed and cadence pick a fitted OpenSim leg cycle by stride length, and each arm swings by its measured amplitude. A tint marks the arm that swings less, the MDS-UPDRS III total and Hoehn and Yahr stage set the trunk stoop, and a 5 Hz hand tremor scales with the rest-tremor items. Wearable inertial measurement unit (IMU) data drive IMU-derived motion, and every view is labeled schematic or IMU-derived. The gait twin explorer, the mechanics engine’s client, plays one reference skeleton and shows, per joint and sensor tier, a joint moment with its confidence, or an abstention when the sensors cannot support one.

4.3 Evidence Workspace

Selecting a participant lists that person’s anchors, and a second selector marks months with laterality or motor-change outputs. The striatal view places left and right putamen binding beside licensed NIH reference brain geometry and highlights the lower-binding putamen. Absolute binding, signed imbalance, and per-side motor sums stay visible together, and fixed rules label the pair contralateral agreement, near-symmetry, or disagreement, never a mechanism. Assays appear as dated context, and a provenance panel traces every value to its source version and row. A source-history view lists a participant’s scan records as dated binding measurements with clinical, gait and assay observations on their own dates.

The display separates uncertainty attached to an individual prediction, validation findings from the source study, and whether the output exists. The body is a Rajagopal geometry [19] scaled to one reference subject and the brain is the NIH 3D Human Reference Atlas; neither is patient anatomy. Each claim carries a status (measured, limited, unavailable, not established, not evaluated).

5 Results and Evaluation

5.1 Participant Overlap and Temporal Alignment

Of 86 portal participants, 31 have laterality outputs, 42 have motor-change predictions, and 30 have both. Within those 30, only three have both for the same analysis month. The other 27 have outputs for different months, and both dates stay visible. Participant overlap therefore overstates same-month availability. Matching months does not establish simultaneous acquisition.

Seven laterality participants have gait in the 12 months before their laterality month. A separate observation view shows 75 scan records from 36 participants within existing anchors, a count of measurements only.

5.2 Verification and Runtime

Across all 997 anchors, no displayed clinical-history observation postdated its anchor. Tests compare returned values with source tables, recompute imaging and motor asymmetry signs, reject participant–anchor mismatches and records dated after the forecast origin, and keep outcomes hidden unless requested; they establish adapter fidelity only. A local single-worker service completed 3,600 requests over 200 keys without error. At concurrency 1, 8, and 32, 95th-percentile retrieval latencies were about 1, 10, and 35 ms.

5.3 Source-Study Findings Shown in the Workspace

The workspace presents the source studies’ validation findings. The motor-state model assigns 99.5 % of 29,366 assessments to one of five states with high confidence and 0.5 % to a boundary between two, separating clear from ambiguous assignments [23]. The pathway profile links lower nigrostriatal and frontostriatal integrity to greater motor burden and higher sensory–visuospatial integrity to better cognition (Fig. 2) [27], cohort-level context for one participant. Signed putamen asymmetry tracks contralateral motor laterality in PPMI ($r = 0.676$) and the independent Systemic Synuclein Sampling Study cohort (S4, $n = 56$, $r = 0.551$). It adds a small held-out gain over non-directional imaging for 12-month side balance at 1,204 landmarks ($\Delta R^2 \approx 0.018$). The study calls that gain uncertain under a bounded link, a model that keeps predicted side balance inside its fixed range [24]. The longitudinal prediction study passed gates for future-data exclusion and participant and site separation. Its internal gain was small ($\Delta R^2 = 0.015$) and received an information-only verdict (below the prespecified threshold). The model failed gates across enrollment eras, and a six-variable version failed transfer to PDBP. The confidence check returned an indeterminate verdict (an abstention). The forecasts carry limited evidence for individual use, and no check quantifies individual uncertainty. In 8,000 stress runs the gates accepted 0.35 % of null findings as positive results [25].

6 Limitations and Next Steps

The present evaluation establishes retrieval fidelity, temporal filtering, and local service performance, and does not raise predictive confidence. The service replays records of historical PPMI encounters.

The next evaluation asks whether users interpret timing, uncertainty, and validation limits more accurately with the workspace than with a flat table of the same evidence. A six-user interpretation study with five examples is prepared and has not been run. A protocol for a two-site wearable study of the gait twin (120 completers) is written, unregistered, and awaiting feasibility review. A longer-term question is how a candidate stimulation change would affect neural activity and movement inside one linked model. The proposed closed loop connects a Kuramoto-type neural oscillator network to a port-Hamiltonian motion

model [4,6]. Descending commands and sensory feedback link the models, and coordinated-reset stimulation enters through an input port of the neural model. A joint inference layer would calibrate both models and keep candidate stimulation within clinician-specified limits [5] (Fig. 2, columns 3–4). Routes include fingertip-glove vibrotactile coordinated-reset stimulation [18,22] and adaptive DBS driven by implanted recordings [16] or guided by scalp electroencephalography or magnetoencephalography [13]. This prototype implements neither route, and both need further modeling, ethics approval, and clinical partners.

7 Conclusion

AI4PD brings participant-specific outputs into one workspace with dates, source records, and validation context. Among 30 participants in both implemented analyses, only three had same-month outputs, so dates must stay explicit. Future work will link implicated pathways and circuits to a knowledge base of candidate interventions, what is available to treat what. Candidates span circuit-directed medication where evidence exists, anti-neuroinflammatory approaches under investigation, and the stimulation routes of Sect. 6, shown to the clinician as evidence for review. Clinical benefit remains to be established.

Acknowledgments. Data used in the preparation of this article was obtained on 2025-01-06 and 2025-12-20 from the Parkinson’s Progression Markers Initiative (PPMI) database (www.ppmi-info.org/access-data-specimens/download-data), RRID:SCR_006431. For up-to-date information on the study, visit www.ppmi-info.org. PPMI, a public-private partnership, is funded by the Michael J. Fox Foundation for Parkinson’s Research and funding partners, including Aligning Science Across Parkinson’s, AbbVie, Alamar Biosciences, Arrowhead Pharmaceuticals, Arvinas, AskBio, Bial, BioArctic, Biohaven, BlueRock Therapeutics, Bristol Myers Squibb, Calico Life Sciences, Capsida Biotherapeutics, the Critical Path Institute, DaCapo Brainscience, Denali Therapeutics, the Edmond J. Safra Foundation, Gain Therapeutics, GE Healthcare, Genentech, GSK, Insitro, Johnson & Johnson, Eli Lilly and Company, Lundbeck, Merck, Neumora Therapeutics, Neuron23, Novartis, Olink, Regeneron, Roche, Sanofi, Tenvie Therapeutics, UCB, VanquaBio, Voyager Therapeutics, and the Weston Family Foundation. Wearable gait recordings are from the WearGait-PD dataset (Synapse doi:10.7303/syn52540892, CC BY 4.0) [2]. Reference inverse-dynamics checks used the public AddBiomechanics dataset [28] (CC BY 4.0); the portal’s striatal reference anatomy is the NIH 3D Human Reference Atlas brain mesh (3DPX-020960, CC BY 4.0) and its body model the Rajagopal full-body OpenSim model [19,20]. Computing resources were provided by the Texas Advanced Computing Center (TACC) at The University of Texas at Austin.

Figs. 1 and 2 are rendered from the MNI152 2009 nonlinear template [9], the Harvard-Oxford atlases distributed with FSL (Desikan et al. [7] and colleagues), and the Pauli 2018 subcortical nuclei [17] with Nilearn [1] and PyVista [21], and from the Rajagopal OpenSim skeleton [19]; pathway tubes are schematic, no participant image is shown.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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